# SMART LIBRARY REQUEST WORKFLOW IN SERVICENOW

## Introduction

Many libraries still manage book borrowing manually .Students cannot easily know book availability. Librarians face difficulty tracking borrow requests .ServiceNow can automate this process using workflows and approvals.

## 1. Identification Phase

## Introduction

The identification phase is the first step in the project development process. In this phase, the main problem or need for the system is identified. It helps in understanding the current issues in the existing system and finding a suitable solution.

In many educational institutions, library book management is still done manually. This creates several challenges such as difficulty in tracking book availability, managing borrow requests, and maintaining proper records of issued and returned books.

Students often face problems in knowing whether a book is available in the library. At the same time, librarians must manually check requests and update book records, which takes more time and increases the chances of errors.

To solve these problems, the **Smart Library Request Workflow in ServiceNow** project is proposed. This system automates the process of requesting and issuing books by using workflows, approvals, and role-based access control.

The system allows students to submit book requests easily while librarians can review, approve, or reject the requests. By implementing this solution in ServiceNow, the library management process becomes faster, more accurate, and easier to maintain.

## 2. Requirement Analysis

### Introduction

Requirement analysis is an important stage in system development where the requirements of the system are identified and documented. It helps developers understand what features the system should provide and how the system should function.

In the **Smart Library Request Workflow in ServiceNow** project, requirement analysis focuses on managing book requests, tracking book availability, and automating the approval process between students and librarians.

### Functional Requirements

Functional requirements describe what the system should perform.

- Students should be able to request books from the library system.
- Librarians should be able to approve or reject borrow requests.
- The system should automatically update the status of books when they are issued or returned.
- Students should be able to track the status of their borrow requests.
- The system should generate reports such as **Most Borrowed Books**.

### Non-Functional Requirements

Non-functional requirements describe how the system should perform.

- The system should be reliable and secure.
- It should provide role-based access control.
- The interface should be simple and user-friendly.

- The system should provide accurate and real-time updates about book availability.

**Hardware Requirements**

- Computer or Laptop
- Minimum 4GB RAM
- Internet Connection

**Software Requirements**

- Operating System: Windows / Linux
- Platform: ServiceNow
- Browser: Google Chrome / Microsoft Edge

## 3. Project Planning Phase

### Introduction

Project planning is the phase where the activities required to complete the project are organized and scheduled. Proper planning helps ensure that the project is completed successfully within the given time.

In this project, planning includes defining system roles, creating tables, configuring workflows, and testing the system.

### Project Goals

The main goal of this project is to automate the library book borrowing process using ServiceNow.

The system aims to reduce manual work, improve accuracy in tracking books, and provide an efficient platform for students and librarians to manage library requests.

### Project Milestones

The project is divided into several milestones:

- Milestone 1 – Create roles (Student and Librarian)
- Milestone 2 – Create tables (Book and Borrow Request)
- Milestone 3 – Add fields to tables
- Milestone 4 – Configure Flow Designer for approval process
- Milestone 5 – Implement access control and UI policies
- Milestone 6 – Create reports and test the system

### Project Timeline

Week 1 – Problem identification and requirement analysis
Week 2 – Planning and system design
Week 3 – Table creation and field configuration

Week 4 – Flow Designer automation setup
Week 5 – Access control and report creation
Week 6 – Testing, documentation, and demonstration

**Risk Management**

Possible risks include configuration errors, incorrect data entry, and workflow issues. These risks can be minimized through proper testing and validation before deployment.

## 4. Development Phase

### Introduction

The development phase involves implementing the planned features and building the system using the ServiceNow platform.

### Table Creation

Two custom tables are created:

- **Book Table** – stores book details such as title, author, ISBN, and status.
- **Borrow Request Table** – stores request details such as requested by, request date, and status.

### Field Configuration

Fields such as title, author, ISBN, request date, and status are added to the tables.

Status options include **Available, Issued, Lost, Requested, Approved, Rejected, and Returned**.

### Workflow Automation

Using **Flow Designer**, the system automatically sends borrow requests to the librarian for approval.

If approved:

- Book status changes to **Issued**
- Borrow request status changes to **Approved**
- Email notification is sent to the student.

## 5. Project Design Phase

### Introduction

The project design phase focuses on designing the structure and workflow of the system. It includes database design, system architecture, and user interface design.

### System Design

The system is designed with two roles:

- **Student** – requests books
- **Librarian** – manages books and approvals

### Database Design

The system uses two main tables:

- Book table
- Borrow Request table

These tables are connected using reference fields to maintain relationships between book details and requests.

### User Interface Design

The user interface is designed to be simple and easy to use. Students can submit requests through forms, while librarians can view and manage requests through the ServiceNow platform.

## 6. Project Documentation

### Introduction

Project documentation provides detailed information about how the system was designed and implemented.

It includes explanations of system configuration, workflows, tables, and reports used in the project.

### Purpose of Documentation

The purpose of documentation is to clearly explain the objectives, development process, and system functionality.

Documentation helps faculty members evaluate the project and helps developers understand the system in the future.

### Technical Documentation

Technical documentation explains the tools and technologies used in the project, such as ServiceNow features including Flow Designer, UI Policies, and Access Control Rules.

### User Documentation

User documentation provides instructions for system users on how to perform tasks such as submitting borrow requests, approving requests, and generating reports.

## 7. Project Demonstration

### Introduction

The project demonstration phase presents the working of the system to show how the developed solution operates in real-time.

### Demonstration Process

During the demonstration:

1. A student logs into the system and submits a borrow request for a book.
2. The request is sent to the librarian for approval.
3. The librarian reviews and approves the request.
4. The system automatically updates the book status to **Issued**.
5. The student receives a notification and can track the request status.

### Outcome

The demonstration shows that the system successfully automates the library borrowing process, reduces manual effort, and improves efficiency in managing library resources.

# STEPS FOR IMPLEMENTATIONS

**1. Create Roles**

**First, create two roles in ServiceNow:**

- **Student**
- **Librarian**

**These roles help control access in the system.**

**2. Create Tables**

**Create two custom tables:**

- **Book Table (u_book) – stores book information**
- **Borrow Request Table (u_borrow_request) – stores book request details**

**3. Add Fields to Book Table**

**Add the following fields:**

- **Title – String**
- **Author – String**
- **ISBN – String**
- **Status – Choice**

**Status values:**

- **Available (Default)**
- **Issued**
- **Lost**

**4. Add Fields to Borrow Request Table**

**Add the following fields:**

- **Requested By – Reference to User**
- **Book – Reference to Book table**
- **Request Date – Date/Time**
- **Status – Choice**

**Status values:**

- **Requested (Default)**
- **Approved**
- **Rejected**
- **Returned**

**5. Configure Reference Qualifier**

**Add a Reference Qualifier on the Book field in Borrow Request table.**

**Condition:**
**Status = Available**

**This ensures students can only select available books.**

**6. Create Flow in Flow Designer**

**Create a flow called Borrow Request Approval Flow.**

**Steps:**

- **Trigger when Borrow Request is created or updated.**
- **Send approval request to Librarian.**

**If approved:**

- **Update Book Status → Issued**
- **Update Borrow Request Status → Approved**

- **Send email notification to student.**

**If rejected:**

- **Update status → Rejected.**

**7. Create UI Policy**

**Create a UI Policy for Borrow Request table.**

**Condition:**

- **When Status = Issued**

**Action:**

- **Make Return Date field mandatory.**

**8. Configure Access Control (ACL)**

**Set role-based permissions.**

**Book Table**

- **Students → Read**
- **Librarians → Read, Write, Delete**

**Borrow Request Table**

- **Students → Create, Read**
- **Librarians → Read, Write, Delete**

**9. Create Report**

**Create a report called Most Borrowed Books.**

**Settings:**

- **Source Table → Borrow Request**
- **Group By → Book**
- **Aggregate → Count**

**This shows the most frequently borrowed books.**

**10. Testing**

**Test the system with sample data:**

1. **Student submits borrow request.**
2. **Librarian receives approval request.**
3. **Librarian approves request.**
4. **Book status changes to Issued.**
5. **Student receives email notification.**